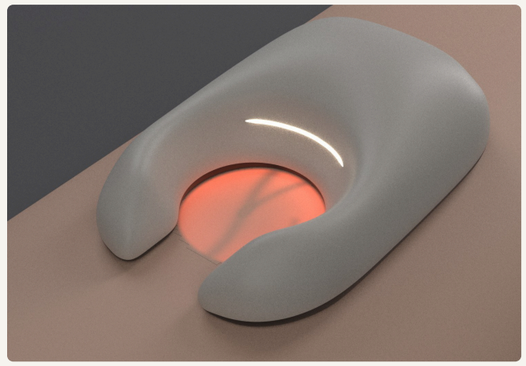
A river stone that lights the skin from within

Palm-sized, naked-eye transillumination vein finder. No camera, no screen, no app — the skin inside the crescent is the display.

PERSONAL PROTOTYPE FOR BENCH EVALUATION — NOT A CLEARED MEDICAL DEVICE

Gates: ● **Concept** ○ CAD ○ Electronics ○ Firmware ○ Verification

01 Rationale



In use (render): red-lit skin in the window, veins as dark lines; status seam on.

How it works

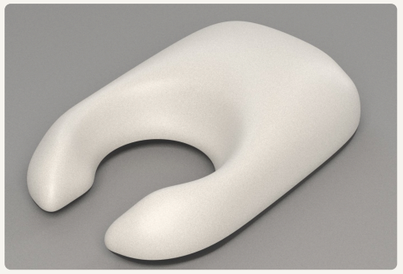
Twelve red LEDs sit in black pockets just outside a 30 mm crescent window and push light *into* the skin under a narrow lip. It scatters through the tissue and glows back out inside the window; blood absorbs it, so veins read as dark lines. Nothing shines toward the eye.

Two banks, visible light only. 660 nm deep red reaches deeper and is absorbed less by melanin (darker skin, deeper veins); 620 nm red-orange looks ~5× brighter to the eye for superficial veins.

Design moves

1. **One G2 skin.** The whole stone is a single cubic B-spline surface — curvature-continuous by construction, no fillets, no flats (its two parametric poles are G1 today; made exact at the CAD gate). The only edge is the waterline.
2. **The waterline.** One planar parting line low on the flank splits a warm-grey PETG top from a black, light-tight optical sole. Both print outer-surface-up, no supports.
3. **The seal is the form.** A gently concave sole (R120) saddles the forearm; the index finger lands just behind the window, where pressing seals the lip.
4. **Hidden controls.** A 0.5 mm finger dimple is the button (flexing membrane). A hairline crescent on the rim glows through 0.6 mm PETG for status and charging. USB-C sits on the waterline at the tail.

02 Three forms, one envelope — same 30 mm window, 16 mm gap, waterline and sole



A — Pebble CHOSEN

One convex dome, the crescent bored through its nose. 90.4 × 58.9 × 25.4 mm.

✓ 600 mAh cell fits · 1.9 mm · ✓ switch fits · 0.6 mm · ✓ support-free · ✓ convex everywhere

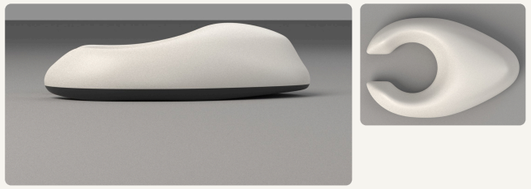


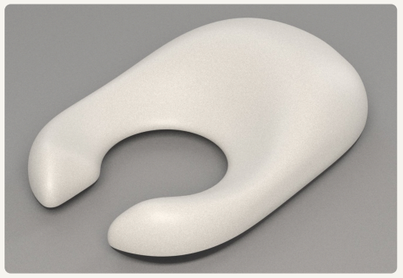


B — Crescent

Teardrop; crown on the window rim; horns taper to soft points. The most dynamic.

✕ cell does not fit the tapering tail · ✓ switch fits · ✓ support-free

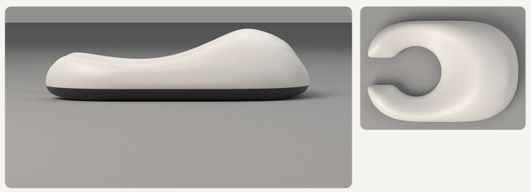




C — Hagstone

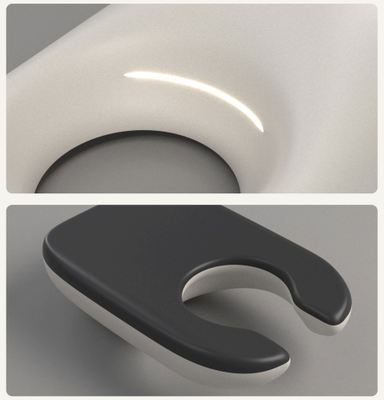
Flat lens with a worn crater; the opening is the hero. Lowest at 23 mm.

✕ no height for cell or switch · ✕ funnel lets in room light · ✓ support-free



03 Chosen: A — Pebble

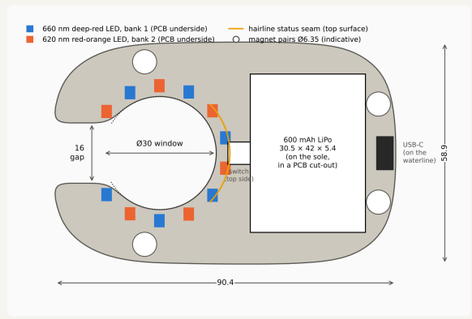


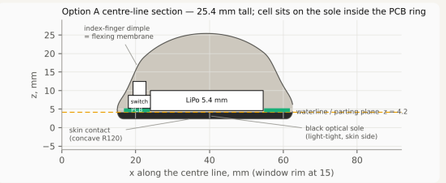


90×59×25	≈ 55 g	2.7–5.8 h	23–50 mA	≈ $56–70
mm (brief ~90×58×26)	estimate (limit 80 g)	full brightness, 600 mAh (need 45 min)	per LED, capped by skin ≤ 41 °C	one unit incl. PCB + shipping (≤ $75)

Why A

- **Only A packages.** The 30.5 × 42 × 5.4 mm cell lies across the crown, on the sole inside a PCB cut-out; the switch sits under the dimple just ahead of it. B's tail and C's crater cannot hold them.
- **It reads as one stone.** A single convex dome: the horns taper out of the body instead of growing from it, so it never reads as a horseshoe.
- **It holds like a pebble.** Thumb and middle finger pinch the tall mid-flanks; the index presses just behind the window, where the seal needs force. Symmetric for either hand.





04 Where the brief meets physics, cost or printing — and the smallest change

LEDs "angled 30–45°"
A flat PCB can't tilt SMD LEDs, and in skin light forgets its direction within ~0.5 mm.
→ Flat LEDs facing the skin, in pockets behind a 2.5 mm lip (r ≈ 17.5): 5× more rim light than at r = 20.

Snap-fit vs. support-free
Both halves print waterline-down, so nothing can cross the waterline.
→ Three PETG clips printed flat (strongest layers) + 4 magnet pairs. No visible screws.

30 mm window
Diffuse light falls ~e-fold per 2.5 mm: a bright 6–8 mm band at the rim, dim centre in room light.
→ Keep 30 mm (needle room); vein near a side rim. Radius is one CAD parameter (24 mm ⇒ centre 5–7× brighter).

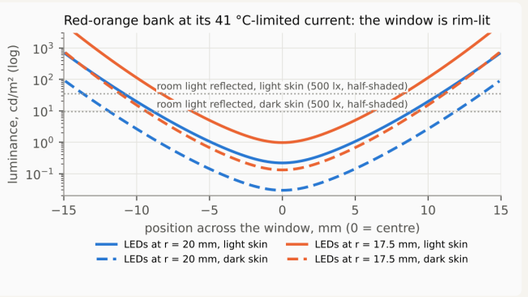
Height ~26 mm
A lower stone looked better but missed the cell by 1 mm.
→ 25.4 mm, set by cell + switch.

Brightness vs. battery
Skin heating under each LED, not energy, sets the ceiling (≤ 41 °C ⇒ 23–50 mA per LED).
→ 600 mAh cell (850–1000 mAh don't fit): 2.7–5.8 h. Firmware cap + NTC at the hottest LED.

Concave seal
Forearms are R 35–50; the back of the hand and the elbow crease are flatter.
→ R120 across, rails 44 mm apart; seals under light pressure. A bench-tunable parameter.

Walls ≥ 1.6 mm
The glowing seam and the flexing button both need thin PETG.
→ Two local 0.6–0.8 mm zones backed by ribs; all other walls ≥ 1.8 mm.

G2 everywhere
The USB-C opening and the waterline must be edges.
→ They are the only two; the port gets a small round.



Form: G2 surface engine in build123d/OCCT — one periodic C2 B-spline skin, fit ≤ 0.12 mm. Renders: Mitsuba 3 path tracer. Budgets: diffusion (Farrell 1992) and hemispherical-source heating; tissue parameters are literature ranges, not measurements.
Evidence: docs/01-concept/concept.md.

Parts & prices: distributor listings via search index (direct site access blocked) — re-verified at the electronics gate.